Stop the Fire!

A Network Optimization Approach to Wildfire Containment

IE 492 — Senior Design Project · Final Presentation

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Real-world graph

7 strategies

Decision support

MOTIVATION

Why now

Reactive → Proactive

Current wildfire response is reactive — we propose strategic, graph-theoretic decision-making before the front hits a village. Stop the Fire! is a concrete prototype for the Turkish Mediterranean belt — the Köyceğiz–Marmaris region.

- Environmental
- Human cost
- Economic damage

CRITERION 1: PROBLEM

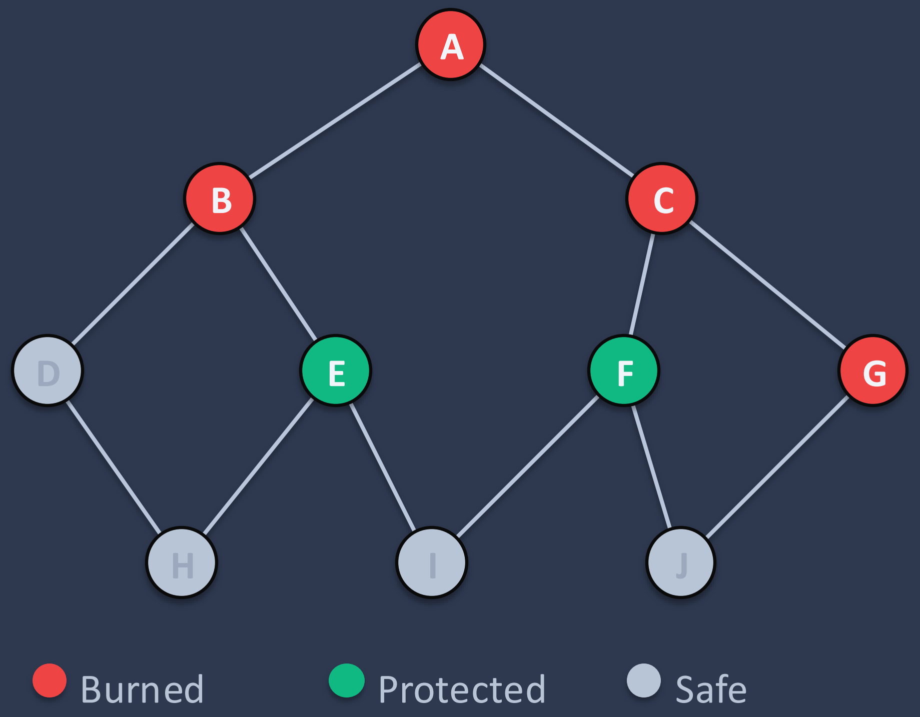
The Firefighter Problem

FORMAL STATEMENT

"Given a graph G, a fire origin, and k resources per turn, minimize total burned vertices."

- Classical combinatorial optimization · NP-Hard
- Deterministic network optimization with limited intervention
- Our adaptation: planar / map-based graphs on real geography

TOY EXAMPLE



DESIGN EVOLUTION

From midterm to now

"We made promises at midterm — here's what we delivered, across five axes."

Axis	Midterm (April)	Final (May)
Strategies	4	7 + 1 Random (+ 3 since midterm)
Network	n=30 synthetic random	Real Köyceğiz map (ESA WorldCover + Lloyd's)
Testing	30v × 15 runs	4,752 synthetic sims + real-map runs
Deliverable	sim engine	Web suite (FastAPI + React)

SYSTEM

System pipeline

Four stages — bbox in, decision out.

01

Real Map

ESA WorldCover · OpenStreetMap
land cover · villages · rivers

02

Density-Aware Graph

Lloyd's k-means · forest vertices
vertices = nodes · edges = links

03

7 Strategies

Greedy · Structural · Lookahead
run on the graph

04

Decision Support

Recommend best · Web Suite
pick the winner

MAP → GRAPH

Density-aware vertex placement

PICK A REGION — 3 WAYS

01 Preset	02 Shift + drag	03 Type coords

01 Forest mask

- ESA WorldCover — open dataset
- It labels every 10 m square: forest, shrub, crop, water, built-up
- Demo: cell = forest if ≥ 55% trees (more than half)
- Not fixed — user can change

02 Lloyd's k-means

- Places vertices on random spots first
- Then moves each to the thick forest near it
- Stops after 14 rounds (Köyceğiz)

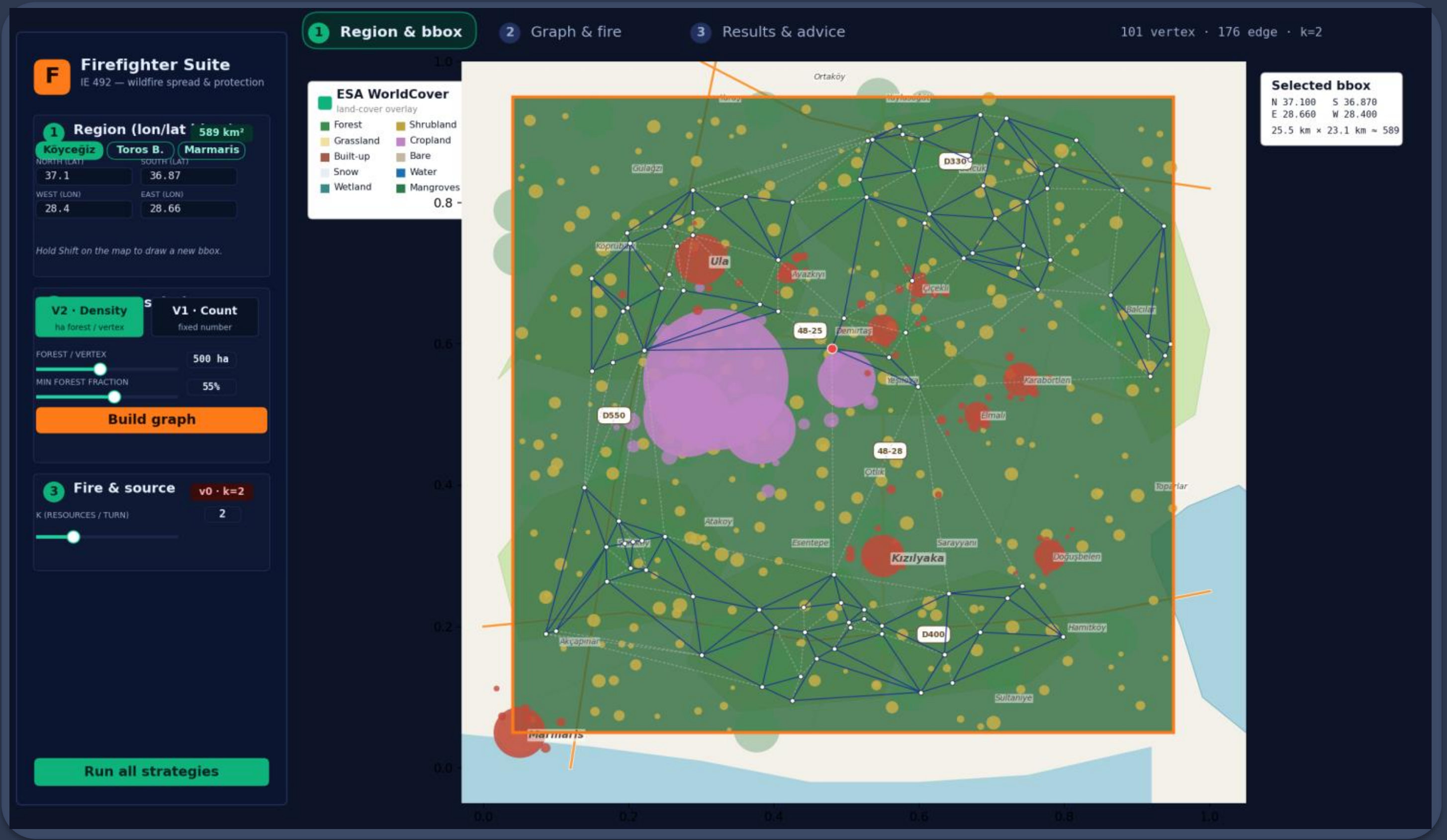
03 Snap to densest

- If a vertex sits on a village or bare land
- Move it to the closest dense forest cell

DENSITY MODE · How many vertices?

Köyceğiz: 500 ha / vertex → 101 vertices

smaller = denser graph · bigger = sparser



Vertices are on the forest — model is for forest fire · next slide's buffer rule excluding obstacle like villages and rivers.

Edge filter — five physical rules

After the vertices, we decide which pairs become edges using five physical rules — if an edge fails any rule, it is blocked.

Bare strip > 60 m

a bare strip longer than 60 metres — fire can't jump it.

Settlement < 600 m

within 600 metres of a village — for safety.

River < 80 m

big rivers — they act as firebreak an 80-metre buffer.

Lake or sea on the path

if the line crosses open water, fire can’t cross it.

Low vegetation density

if the vegetation is too thin, there is not enough fuel for fire.

KÖYCEĞİZ EXAMPLE

278 candidates → 176 active · 102 blocked

About one third removed

STRATEGIES

Strategy portfolio — 3 philosophies

Greedy

LOCAL / FAST

- Max Degree
- ★ **Max White Neighbors**

"What's good now?"

Structural

GLOBAL TOPOLOGY

- Min Cut Edge Front
- ★ **Min Damage Cut**
- Betweenness Front

"Where's the bottleneck?"

Lookahead

FORWARD-LOOKING

- ★ **One Step Lookahead**
- Hybrid Density Aware

"What happens 2 turns later?"

★ = best representative. 7 documented strategies. Top 5 strategies will be given in detail.

STRATEGIES

Lookahead breakthrough — one_step_lookahead

THE IDEA

Don't just ask "what's good now?"

Ask "what happens 2 turns later?"

Top-6 candidates × **C(6,2) = 15 pairs**

Per pair simulate: **protect → spread → greedy → spread**

Pick the pair with **lowest burned count**

RESULT

0.5 ms · 6% relative gain over greedy · #1 overall

PAIR (v1, v2) SIMULATION

Now (t=0) → Protect (v1, v2) → Spread (t=1) → Greedy (k=2) → Final (t=2)

→ REPEAT FOR ALL 15 PAIRS

Pick the pair (v1*, v2*) that minimizes burned at t=2.

0.5 ms | **15 sims** | **k=2 native**

The engine simulates the engine — full k=2 protect-then-spread dynamics inside each candidate evaluation.

DESIGN EXPERIENCE

Reframing Structural Approach

Critical Understanding

" Min Cut Vertex Front is asking the wrong question."

"It says 'don't let fire reach the distant vertex,' but our main objective is 'minimize total burned.'"

OUR REFORMULATION — Min Damage Cut

score = saved / cut_size (ratio, not absolute)

BFS shells	Burned	Best individual case
depth 2 / 3 / 4	**39.4% → 30.3%**	**1 vertex → 12**
tested	(-23% relative)	(12× ratio)

feedback → reformulation → measurable improvement.

RESULTS

Synthetic benchmark — 4,752 simulations

#	Strategy	Burned %	Cost
1	one_step_lookahead ★	**27.5%**	2.2 ms
2	betweenness_front	**38.5%**	17.6 ms
3	max_white_neighbors	**41.5%**	0.3 ms
4	hybrid_density_aware	**42.2%**	36.0 ms
5	max_degree	**42.4%**	0.2 ms
6	min_damage_cut	**42.6%**	50.3 ms
7	min_cut_edge_front	**43.4%**	11.1 ms
8	random	**70.1%**	0.2 ms

TAKEAWAYS

#1 by burn%
one_step_lookahead
27.5% burned

Random gap
43 pts (27.5→70.1%)

Runner-up
betweenness_front 38.5%

4,752 runs = **3** sizes × **18** RNG seeds × **11** starting points / graph × **8** strategies

All results above use the default parameter **k = 2**.

RESULTS

Real-map validation — Köyceğiz Example

Vertices	Active edges	Blocked edges	Forest total	Fire origin	k
101	172	106	50,280 ha	v0 (deg 7)	2

Ranking confirmed on the real map.

LIVE

Live demo — Web Suite

3-CLICK RUN ON STAGE

1. **Köyceğiz preset → Build graph**
   ~1 s; in-memory cache warm

2. **Click v0 (hub) → Run all strategies**
   11 strategies in parallel; ~2 s

3. **Results → read the recommendation card**
   graph-specific numbers, not a template

firefighter-ie492.netlify.app/

Köyceğiz preset

CLOSING

Conclusions & future work

Validated ✓

- 4 → 7 strategies (3 philosophies)
- Real-map deployment (Köyceğiz)
- Working web suite delivered

Findings

- No single dominator — topology + k drive choice
- Lookahead wins at scale; betweenness 2nd on cost
- Reframing (min-cut → min-damage) measurable

Future

- Stochastic spread (wind / elevation / fuel)
- Multi-period budget (carry-over k)
- OGM live integration (real-time fire feed)

Thank you — Questions?